

American beech sapling dominance: Stand-scale insights from mobile laser scanning

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ABSTRACT

For several decades, the proportion of American beech (*Fagus grandifolia*) has increased in sugar maple (*Acer saccharum*) stands in North America, particularly in the understory, where it can hinder the regeneration of other species. Although this dominance is well documented, the underlying mechanisms by which it develops remain poorly understood. We aimed to determine whether the spatial relationships between beech saplings, mature beech trees, and canopy openness, previously observed among stands using discontinuous plot-based sampling, also applies within stands using continuous sampling of full stands. Our objective was to better understand how this dominance develops, as it is uncertain whether these relationships hold at finer spatial scales within stands, i. e., whether beech dominance within a stand is spatially related to canopy gaps and/or mature beech density. We created 3D-maps of 11 1-ha stands in Québec, Canada using mobile laser scanning (MLS) technology and developed a new method for analyzing regeneration which allowed us to map a total of 8455 trees and 30,498 saplings and to investigate the relationship between beech saplings and mature beech trees as well as canopy openness at various spatial scales. Contrary to our expectations, we found very little to no relationship between beech saplings and canopy openness, and a highly variable relationship with mature beech trees among sites. Our findings underscore the importance of spatial scale in regeneration studies and suggest that beech proliferation may be difficult to reverse, as sexual reproduction appears to play a more important role in beech regeneration than previously thought.

1. Introduction

For several decades, the proportion of American beech (*Fagus grandifolia* Ehrh.) in northeastern temperate forests has increased significantly, particularly in the understory (Bose et al., 2017). For example, Duchesne et al. (2005) determined that the proportion of beech saplings within permanent plots had doubled over a period of just 10 years in Québec while Bose et al. (2017) established that beech-dominated stands occupied 27,4% of all northeastern USA forest area, an approximate 5% increase over the past 16 years. During this same period, a decline in health (Horsley et al., 2002) as well as a decrease in the productivity (Nolet and Kneeshaw, 2018) of sugar maple (*Acer saccharum* Marsh.), a key species with high commercial value, has been observed in northeastern North American temperate forests, particularly in Québec (Duchesne et al., 2005).

American beech is a highly shade-tolerant, late-successional species that typically establishes and persists under closed canopies on a wide range of mesic, well-drained soils, including loamy and fertile sites,

where it can maintain a persistent regeneration layer while suppressed in low-light understories (Burns and Honkala, 1990; Tubbs and Houston, 1990). Sugar maple is also a shade-tolerant, late-successional species, but its regeneration is more strongly linked to mesic, nutrient-rich, well-drained soils, particularly calcium-rich and base-saturated substrates that support optimal seedling establishment and growth (Burns and Honkala, 1990; Godman et al., 1990). Although capable of long-term understory persistence, its recruitment and early growth are more sensitive to edaphic conditions and microsite quality compared to beech (Beaudet et al., 1999).

The proliferation and expansion of beech at the expense of sugar maple is problematic from an economic standpoint, as the commercial value of sugar maple is significantly higher than that of American beech. The commercial value of American beech is also further diminished by beech bark disease (BBD), which limits the growth and degrades wood quality of infected trees (Kasson and Livingston, 2012). Beech proliferation in combination with BBD is also problematic from an ecological perspective as they lead to changes in regeneration dynamics (Nyland

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et al., 2006), in biodiversity (Cale et al., 2017), as well as changes in the structure and composition of forests (Duchesne et al., 2005). These transformations underscore the multifaceted challenges posed by the increasing dominance of American beech in northeastern temperate forests.

Multiple factors contributing to beech dominance in the understory have been previously identified. American beech's greater tolerance to acidic soils and poor growing conditions is a factor suggested by various studies (Duchesne and Ouimet, 2009; Kobe et al., 2002). Another potential factor is beech's greater ability to grow in the understory of stands due to its high shade tolerance (Canham et al., 1988). Following its establishment in deeply shaded understories, the creation of openings in the forest canopy through various disturbances such as partial cutting can lead to the proliferation of beech (Bose et al., 2017; Nolet et al., 2008). Beech can reproduce rapidly through root suckering following injuries, which can be caused by logging machinery (Houston, 2001). Beaudet et al. (2007) found that shoots originating from root suckering are more vigorous than those originating from sexual reproduction and that a high proportion of these suckers could explain the dominance of beech in the understory over maple. However, distinguishing between root suckers and stems originating from sexual reproduction, either through root excavation or genetic analysis is a difficult, time-consuming, and costly task, making it impractical to apply at the scale of an entire 1 ha stand or even a single plot (e.g., 400 m²). Hence, it is very difficult to establish the respective roles of vegetative (root suckers) and sexual regeneration in beech sapling proliferation exacerbating the difficulty of understanding this phenomenon. Some studies also mention the possibility that beech bark disease influences its establishment in the understory by creating gaps and promoting reproduction through root suckering, although this idea is questioned (Roy and Nolet, 2018). Finally, another potential factor is selective herbivory, particularly by ungulates, which prefer to consume sugar maple and yellow birch (*Betula alleghaniensis* Britton.) seedlings over American beech seedlings (Bose et al., 2018; Rossell et al., 2007).

In short, the dominance of beech in the understory has long been documented and studied from various perspectives. Yet, the specific mechanisms by which this dominance develops within a given stand remain unclear. At the landscape level, it has been observed that more open stands, especially those containing mature beech trees, are more likely to develop a dense layer of beech saplings (Nolet et al., 2015). These relationships have been determined using standard forest sampling methods, which rely on separate, non-contiguous plots across multiple stands. However, it is uncertain whether these relationships hold true at finer spatial scales within stands, i.e., we do not know whether beech dominance within a stand is spatially related with canopy gaps, and/or mature beech density. Here, we aimed to determine whether the spatial relationships between beech saplings, mature beech trees, and canopy openness, previously observed among stands using discontinuous plot-based sampling, also applies within stands using continuous sampling of full stands. Understanding these relationships is crucial for informing silvicultural practices.

In this study, our main objective was to better understand at a fine scale how beech dominance develops in the understory within sugar maple-dominated stands and to address scale-related research questions concerning spatial relationships between beech saplings, mature beech trees, and canopy openness. Because beech saplings are assumed to develop mainly through root suckering and that they respond strongly to increased light availability, we expected that (1) beech sapling cover would be positively related to mature beech basal area and canopy openness, and (2) the relationship between beech sapling cover and canopy openness would be strengthened by higher beech basal area, the rational being that without beech parents, there would be no beech saplings no matter the canopy openness. These expectations, (i.e., beech basal area and total basal area having opposite effects on beech saplings) are consistent with what has been observed at larger scales. To test our hypotheses, we created 3D-maps of eleven (11) 1-ha stands using mobile

laser scanning (MLS) technology which allowed us to map a total of 8455 trees and 30 498 saplings and to investigate the relationship between beech saplings and mature beech trees as well as canopy openness at various spatial scales.

2. Materials and methods

2.1. Study area

The study took place in two separate clusters within the Outaouais region (Québec, Canada). Half of the sites (5 out of 11) were located northeast of Duhamel, close to Gagnon Lake (46°06'36"N, 75°08'55"W), which is in the western sugar maple – yellow birch bioclimatic region (Morneau, 2021). The mean annual temperature is 4.5 °C and the mean annual precipitation is 1091.1 mm (with 201.4 mm as snow precipitation). The average number of degree-days above 0 °C was 2804 and the average number of days with minimum temperature > 0 °C was 174.4 in between 1981 and 2010 (Environnement Canada, 2024). The study area's surface geology primarily consists of thin to moderately thin glacial till, predominantly made up of metamorphic rocks like gneiss. This terrain is characterized by numerous hills, with elevations generally below 350 m above sea level (a.s.l.) (Robitaille and Saucier, 1998). Overlaying the parent material is a layer of sandy Dystric Brunisols (USDA classification: Typic Dystrichrepts) (Soil Landscapes of Canada Working Group, 2010). The forest canopy is dominated by sugar maple in association with American beech, yellow birch, American bass-wood (*Tilia americana* L.), ironwood or American hop-hornbeam (*Ostrya virginiana* (Mill.) K. Koch), eastern hemlock (*Tsuga canadensis* (L.) Carrière), and balsam fir (*Abies balsamea* (L.) Mill.). In this cluster of sites, canopy harvesting treatments were implemented in autumn 2006 and consisted primarily of selection cutting (~30% basal area removal across all diameter classes), with untreated controls and nearby clear-cuts embedded in the broader treatment matrix; all sampled sites were selected to exclude clear-cuts and therefore represent a mixture of selection-cut and control patches. The most recent perturbation for all these sites is the mortality induced by BBD.

The other half of the sites are in the Kenauk forest (45°48'05"N, 74°49'44"W), a private forest territory that spans 265 km². This area belongs to two bioclimatic domains: sugar maple-American basswood and sugar maple-yellow birch (Morneau, 2021). The mean annual temperature is 4.8 °C and the mean annual precipitation is 985.5 mm (with 178.1 mm as snow precipitation). The average number of degree-days above 0 °C was 2888 and the average number of days with minimum temperature > 0 °C was 181.4 in between 1981 and 2010 (Environnement Canada, 2024). The forest canopy varies but is generally dominated by sugar maple, yellow birch, American beech, American basswood, red oak (*Quercus rubra* L.), ironwood, and balsam fir. The terrain features diverse elevations, including hills and valleys, with altitudes ranging from 200 to 300 m. In this area, the predominant soils are well-drained Dystric Brunisols with moder-type humus and a loamy texture formed primarily on glacial till deposits consisting mainly of gneiss, quartzite, and granite (Soil Landscapes of Canada Working Group, 2010). Study sites in Kenauk differed in their management history and canopy disturbance regimes. Most sites were uneven-aged stands treated by selection cutting (Site 2: ~35 years; Site 13: ~20 years), alongside one unmanaged old-growth stand (Site 5), one transitional site at the boundary between an unmanaged stand and a selection-cut stand (~35 years; Site 6), and one site with uncertain history but likely originating from a commercial clearcut (>50 years; Site 4). The most recent perturbation for all these sites is also the mortality induced by BBD.

2.2. General approach and rationale

To test our hypothesis, we needed to develop an improved method for quantifying regeneration, enabling us to examine beech dominance

in the understory distinctly from traditional field inventory methods such as fragmentary microplots used in previous research. Quantifying and studying regeneration into a forest inventory, both at large and fine scale, is a challenging task that involves a high degree of uncertainty while being very labor-intensive. By using MLS to collect data points, and by developing custom algorithms to process them, we were able to map the sapling cover, the sapling density and the sapling count of whole hectares of dense understory forests while also creating a full 3D-map database where all tree stems (diameter at breast height, DBH ≥ 9 cm) were identified and geolocated with their respective DBH and basal area (BA) calculated. This amount of regeneration data would be impossible to gather on the field with traditional sampling methods given the amount of time and effort needed to measure all these saplings. The LiDAR data also permitted us to conduct statistical analysis of all our variables at different scales. We were able to measure beech sapling cover within pixels of varying sizes (ranging from 2×2 m to 16×16 m) and the basal area of beech within plots of varying radii (10, 20 and 30 m) centered around those pixels. These plot sizes represented the largest possible subdivisions that allowed us to retain a sufficient number of replicates within stands (≤ 31) and to run statistical analyses with enough statistical power. All variables require a varying amount of data processing to be extracted from the raw LiDAR data, which is elaborated on later in [Section 2.4](#). To our knowledge, this study is the first to use MLS to quantify forest regeneration in this manner, offering a novel perspective for ecological research.

2.3. Experimental design and forest inventory

This study was conducted across 11 forest sites. At each site, 1-ha plots were established using high-precision geode GPS to define four corners. Paths approximately 7 m apart were cleared to facilitate LiDAR scanning and minimize data loss caused by occlusion from trees or saplings. All beech trees (DBH ≥ 15 cm) and dead trees (DBH ≥ 9 cm) within the plots were marked with flags. The 15 cm DBH threshold for beech trees, rather than the conventional 9 cm limit used to calculate BA, was selected to focus on mature individuals more likely to have influenced the sapling layer compared to younger trees and to limit the number of trees that required field identification. Observation points were created every 7 m along these paths, totaling 180–200 points per plot, where panoramic 360° photos were taken to aid in identifying flagged trees in the LiDAR point cloud. At each observation point, visual estimates of beech sapling relative cover, in relation to all sapling species, were recorded within a 3.5 m radius by the same observer, using 5% increments. Although this approach is semi-quantitative and may be subject to observer bias, it provided a practical and consistent method across all sites and sampling points and avoided the need for exhaustive counts of $> 30,400$ saplings. Visual estimates were cross checked in the field to ensure consistency among observations. While some uncertainty remains, we are confident that estimates were generally within 5 % of actual beech sapling cover, and this level of uncertainty is unlikely to affect the overall patterns or conclusions of the study. Beech saplings were not distinguished by regeneration origin (sexual versus vegetative), and all analyses therefore considered total beech sapling regeneration regardless of origin, as this distinction was not feasible at the spatial extent and sample size of the study, given that excavating the roots of thousands of saplings or performing genetic analyses on all individuals would have been impractical. LiDAR scanning was performed using a VLP-16 (PUCK™ by Velodyne) sensor mounted on a backpack above the user's head to avoid blocking the laser beam. A synchronized voice recording was used to mark observation points along the scanned paths by matching timestamps with LiDAR recordings. The total distance scanned at each site ranged from 1500 to 1800 m.

2.4. LiDAR data processing

The initial LiDAR data is unstructured and consists of a series of

spatially disorganized frames (a frame being a picture of points at a given time). Data acquisition began with the Velodyne LiDAR system recording raw data in a pcap file format, which was stored on a mobile tablet. This pcap file was subsequently converted into a point cloud file (referred to as an osee file) using the Augmented LiDAR Box (ALB) software developed by Outsight®. To transform the individual frames contained in the osee file into a cohesive 3D point cloud, facilitating the identification of geometric features, we utilized the Simultaneous Localization and Mapping (SLAM) solution developed by Outsight®. The SLAM algorithm also enabled the reconstruction of the spatial trajectory followed by the LiDAR sensor during field sampling. This reconstructed trajectory was later employed to map the observation points and panoramic photo locations.

The LiDAR data processing workflow, in combination with field sampling species identification, aimed to derive the following variables: the total basal area (living and dead), the basal area in mature beeches (living and dead), the basal area in beech saplings and the cover created by beech saplings. The steps involved in deriving these variables from the point cloud included both general procedures applicable to all variables and specific methods tailored to individual variables. Additionally, these variables were calculated across multiple spatial scales. For example, basal area calculations were performed for any given location within the stands and at resolutions such as 2 m, 4 m, and 16 m. Most of the LiDAR data processing was conducted using custom Python scripts that were either developed or adapted specifically for this study. These scripts are available upon request to facilitate reproducibility and further research.

2.4.1. Tree segmentation

Segmentation is the creation of objects within the original point cloud, as points are not objects initially. Tree segmentation primarily involved isolating individual tree trunks from other objects within the point clouds, such as branches and leaves. To facilitate this process, we retained only a vertical slice of the point cloud between 0.75 m and 1.75 m above the ground. As conventional cleaning methods did not meet our specific requirements, we developed a customized cleaning approach termed "clean by columns." In this method, the point cloud was first divided into 2 cm-wide vertical columns. These columns were further subdivided into $2 \times 2 \times 2$ cm voxels. Columns were excluded if they did not satisfy specific criteria based on their constituent voxels. To be retained, a column needed to meet the following conditions. First, the column height, measured as the distance between the two furthest voxels containing at least one point, had to be at least 80 cm. Second, at least 10 out of the 50 voxels comprising the column needed to contain at least one point. The "clean by columns" procedure was especially critical due to the presence of dense shrub and sapling layers within the studied stands. Following this cleaning process, a density-based spatial clustering algorithm (DBSCAN, implemented via the scikit-learn library) was applied to the cleaned point cloud to group points into discrete objects ([Ester et al., 1996](#)). Some of these objects were clearly identifiable as mature trees or saplings, while others exhibited less distinct forms.

Each object (except saplings, addressed in [Section 2.4.3](#)) was visually inspected within the point cloud to confirm its classification as a tree. In cases where an object contained multiple trees, it was manually subdivided into the correct number of trees using an in-house Python application developed with PyVista ([Sullivan and Kaszynski, 2019](#)). The DBH of each tree was then estimated using an algorithm that uses the raw data of the LiDAR sensor that avoids the noise produced during the SLAM process ([Nolet and Rochon, 2022](#)). This algorithm allows a very precise estimation of field DBH, even for small stems.

2.4.2. Sapling cover index (and specifically beech sapling cover index)

To calculate the sapling cover of the stand, we started by removing the identified tree trunks (DBH ≥ 9 cm) in the point cloud, leaving sapling trunks along with branches and leaves. To account for the

majority of sapling leaf cover, we retained a vertical slice of the point cloud between 0.75 m and 3 m above the ground. Using the same 2 cm column and voxel approach described previously, we calculated a sapling cover index for the stand. Each column measured 2.25 m in height, consisting of 113 2 cm voxels. Each voxel was assigned a score of 1 if it contained at least one point from the point cloud. A column's total score was then calculated as the sum of its voxel scores, yielding values between 0 and 113. This final dataset, representing column scores, was rasterized in QGIS to generate a localized sapling cover index across the stand. This process associated the number of occupied voxels within a column with the density of saplings in that area. To calculate the beech sapling cover index, a similar raster was generated to represent the percentage of beech saplings across the stand. The beech percentage raster was created using the localized observation points, within 3.5 m-radius plots, each associated with a percentage of beech saplings recorded by the same observer. A Triangulated Irregular Network (TIN) interpolation was applied to these percentages to produce a raster in which each 2 cm column was assigned a value between 0 and 100. The total sapling cover raster (values between 0 and 113) was then multiplied by the beech percentage raster (values between 0 and 100), resulting in a beech sapling cover index raster with values ranging from 0 to 1130.

2.4.3. Sapling basal area (and specifically beech sapling basal area)

To corroborate the beech sapling cover index (a non-conventional measure), we also calculated the basal area of (beech) saplings within the stand. We first identified all the sapling trunks ($0.5 \text{ cm} \leq \text{DBH} < 9 \text{ cm}$) detected in our tree segmentation. We knew from prior experience and from published studies (Kuzelka et al., 2022) that small saplings ($0.5 \text{ cm} \leq \text{DBH} \leq 2 \text{ cm}$) were harder to segment than trees from the point cloud. A first reason for that is that the studied stands had dense shrub layers and foliage, which limited the penetration and accuracy of LiDAR beams in capturing small trunks. A second reason is that the clean by column treatment may fail to detect small DBH saplings and crooked saplings. To further assess segmentation accuracy, we manually identified all visible stems (trees and saplings) within the point cloud of a single site, yielding a total of 3250 individuals. The segmentation algorithm detected 2955 stems, including 1449 saplings, corresponding to an accuracy of 91%.

Furthermore, DBH estimation for saplings may not be as precise as for trees since the arcs (used in DBH estimation) contain a lower number of points (typically one to three), leading to less precise DBH estimates. To address this issue, all detected saplings with a calculated DBH below 2.01 cm were assigned a standardized DBH of 2 cm to avoid underestimating their contribution to basal area. The selected threshold of 2.01 cm reflects our confidence that saplings above this size were consistently detected by the segmentation procedure. Assigning a standardized DBH of 2 cm to smaller detected saplings has a negligible influence on total sapling basal area, as individuals with DBH between 3 and 9 cm contribute most of the basal area, making this adjustment a reasonable correction for undetected smaller saplings. Using this adjustment, the basal area of all identified saplings was computed. This basal area dataset was then combined with the beech percentage raster to derive the beech sapling basal area for the entire stand.

2.4.4. Beech and dead tree identification and verification

To identify mature beech trees and dead trees within the stand, a manual point picking was conducted within the point cloud. This process involved integrating the spatial trajectory followed by the LiDAR sensor during field sampling, field observation points, and panoramic photographs taken during sampling along this trajectory. Flagged trees in the photographs were manually located within the point cloud, and their coordinates were identified using the CloudCompare software. To evaluate the accuracy of our tree detection algorithm, we compared the number of American beech trees identified in the point clouds with those recorded during field sampling in 5 sites. The algorithm detected 626

beech trees with $\text{DBH} \geq 15 \text{ cm}$, representing 97% of all stems and 98% of the total basal area, relative to the 645 stems measured in the field. Most of the undetected individuals were small-diameter trees growing at steep angles to the ground, a known limitation of our segmentation approach. These missing stems were subsequently verified and manually added to the database. The tree segmentation process generated various classifications of segmented trees, many of which required manual verification to ensure no mature trees were overlooked by the algorithm. A common issue involved mature trees being misclassified as large clusters of connected points, labeled by the algorithm as "bushes." All DBH estimates for mature trees were manually reviewed to identify and correct substantial calculation errors. Additionally, all classifications of segmented trees were manually verified to prevent data loss of mature trees.

2.4.5. Scales and variables

Given the extensive scale of the collected field data (a fully sampled hectare for each site), we aimed to analyze the data across multiple spatial scales to investigate the relationships between beech saplings, mature beech trees, and canopy openness, and at which scale these relationships occurred. Each site was divided into localized square pixels, with sizes ranging from $2 \times 2 \text{ m}$ to $16 \times 16 \text{ m}$ in 2 m increments. Various variables were averaged within these pixels to capture localized patterns. Additionally, some of these variables were calculated within circular radii plots centered on these pixels, with radii ranging from 5 m to 30 m, varying in steps depending on the variable. The variables analyzed in those plots included basal area measurements for different species and their conditions, such as beech trees, beech trees over 25 cm in DBH, and all trees, as well as their living, dead, or combined states. This multiscale approach allowed us to compare localized and broader spatial patterns and investigate their influence on sapling cover and other stand-level characteristics. An overview of the variables of this study and their associated spatial scales is provided in Table 1 and Fig. 1.

2.5. Statistical analysis

For each site, beech sapling cover and basal area (two dependent variables) were analysed with linear regressions to examine their relationships with the calculated basal area variables across the scales described above, i.e., from $2 \times 2 \text{ m}$ to $16 \times 16 \text{ m}$ pixels for the dependent variable and 10, 20 and 30 m plots radii for the independent variable. Furthermore, a series of single linear regression analyses were conducted using multiple combinations of the independent variables, according to species composition (all species or beech only), tree status (alive, dead, or combined), and beech DBH thresholds ($>15 \text{ cm}$ or $>25 \text{ cm}$). Considering the 11 sites, eight spatial scales, and numerous explanatory variable combinations, we conducted a total 4752 (11 sites $\times$ 216 combinations of independent variables $\times$ two dependent variables) linear regression models using our dataset (Moore et al., 2025). The interaction between the distribution and abundance of mature beech trees and the degree of canopy openness was also included as an explanatory variable in these models. But given that this interaction did not significantly affect beech sapling cover and basal area, we removed it from the models to adhere to the principle of parsimony. The value of R^2 was used to estimate the relationships quality. In this study, data that originates from a single site are spatially structured on a regular grid, meaning that spatially proximate observations may exhibit similar responses due to shared or overlapping information (Fig. 2). Our analyses are specifically designed to compare the strength of spatial relationships across multiple spatial scales within this same site, using R^2 as the primary metric of comparison. In this context, any underlying spatial autocorrelation would be present consistently across all scales considered, rather than affecting one scale disproportionately. Consequently, while spatial autocorrelation can influence statistical inference (e.g., p-values), it does not alter the relative magnitude of R^2 values across scales. Because our objective is to compare R^2 values across scales rather

Table 1

Description of all variables used in the linear regression analyses. Y: dependent variable; X: independent variable.

Variable	Type (x, y)	Scales	Calculation	Indicator of	Detail
Beech sapling cover index	y	Pixels, 2 m to 16 m (2,4,6,8...)	See Section 2.4.2	Beech sapling cover	NA
Beech sapling basal area	y	Pixels, 2 m to 16 m (2,4,6,8...)	See Section 2.4.2	Beech sapling cover	NA
Living mature beech basal area	x	10, 20 and 30 m radius, centered around the pixels	Sum of the basal area of each living beech tree in radius i	Living mature beech basal area	Includes alive beech trees (DBH > 15 cm)
Dead mature beech basal area	x	10, 20 and 30 m radius, centered around the pixels	Sum of the basal area of each dead beech tree in radius i	Standing dead mature beech basal area	Includes standing beech trees (DBH > 15 cm)
Alive and dead beech basal area	x	10, 20 and 30 m radius, centered around the pixels	Sum of the basal area of each standing beech tree in radius i	Total standing mature beech basal area	Includes all standing beech trees (DBH > 15 cm)
Living tree basal area	x	5, 10 and 15 m radius, centered around the pixels	Sum of the basal area of each living tree in radius i	Canopy closure	Includes living trees (DBH > 9 cm)
Dead tree basal area	x	5, 10 and 15 m radius, centered around the pixels	Sum of the basal area of each dead tree in radius i	Trees to not consider in basal area calculations	Includes dead trees (DBH > 9 cm)
Alive and dead tree basal area	x	5, 10 and 15 m radius, centered around the pixels	Sum of the basal area of each standing tree in radius i	Canopy closure	Includes all standing trees (DBH > 9 cm)

than to perform significance testing, accounting for spatial autocorrelation would not change the interpretation of our results but would instead add unnecessary complexity to the statistical model. Importantly, we visually inspected all possible relationships (more than 4000 plots) to ensure that no systematic nonlinear patterns were present between the dependent and independent variables. We used R 4.2.2 ([R Core Team, 2022](#)) for all the statistical analysis using packages ggplot2 ([Wickham, 2016](#)) and gridExtra ([Auguie, 2017](#)).

2.6. Determining the scale of all analysis

To find at which scale the strongest relationships occurred, we compared the average R^2 of models built at all the different radius at which mature beech basal area was calculated (10, 20 and 30 m) at all the different pixels that the beech sapling cover index was calculated (from 2×2 m to 16×16 m).

2.7. Determining how to quantify mature beech basal area

To verify the relationship between mature beech basal area and sapling cover, we tested multiple ways of quantifying mature beech basal area. We calculated the beech basal area for all standing beech stems over 15 cm, dead and alive. We also calculated the basal area for

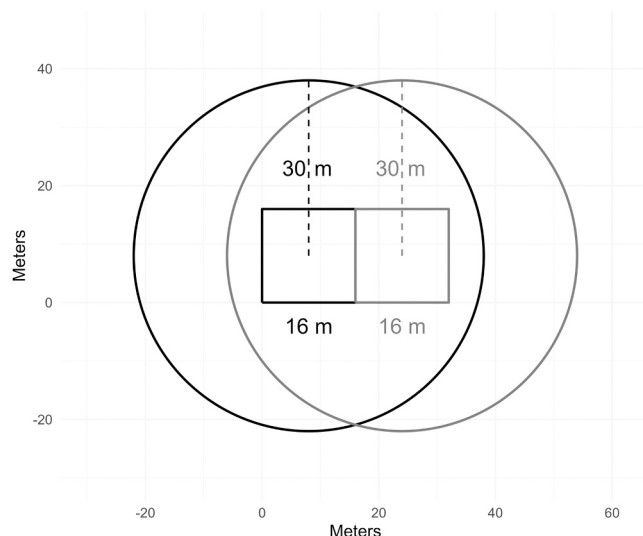


Fig. 1. Illustration of two 16×16 m pixels (used for the calculation of beech sapling cover and basal area variable) with their respective circular 30 m radius plots (used for the mature beech basal area variable). The concept remains the same for all other pixel and plot sizes.

beech stems (DBH ≥ 25 cm) to verify if the relationship became stronger with more mature trees. We found that the overall mean R^2 was highest when using mature beech basal area (≥ 15 cm DBH) compared to more restrictive thresholds. Therefore, this variable was selected for subsequent analyses. No strong relationships were observed between standing dead beech basal area and beech sapling cover. Furthermore, including or excluding standing dead beech basal area in the mature beech basal area variable did not influence the relationship with sapling cover. Standing dead trees were included in the analysis due to recent mortality caused by beech bark disease, which may have influenced sapling regeneration prior to tree death.

3. Results

3.1. Stand structure and dataset summary

A total of 8455 trees and 30,498 saplings were detected and mapped across the 11 study stands. The stand identifiers range from 2 to 14, excluding 3 and 7 ([Table 2](#)). This exclusion was due to SLAM processing challenges with a few scans. The average tree DBH ranged from 16.1 cm to 24.9 cm, with average beech DBH varying between 17.6 cm and 26 cm. Average total basal area ranged from $15.6 \text{ m}^2 \cdot \text{ha}^{-1}$ to $23.2 \text{ m}^2 \cdot \text{ha}^{-1}$, while average beech basal area varied from $1.22 \text{ m}^2 \cdot \text{ha}^{-1}$ to $7.69 \text{ m}^2 \cdot \text{ha}^{-1}$. Dead beech basal area ranged from 0 to $3 \text{ m}^2 \cdot \text{ha}^{-1}$, and beech sapling basal area varied from $1 \text{ m}^2 \cdot \text{ha}^{-1}$ to $5.2 \text{ m}^2 \cdot \text{ha}^{-1}$ ([Table 2](#)).

3.2. Scale effects on spatial relationships in beech regeneration

Across all variables and scales, the strongest relationships were consistently observed at the largest scales ([Fig. 2](#) and [Fig. 3](#)). For example, at the site level, the relationship between beech sapling cover index and mature beech basal area in site 9 increased from an R^2 of 0.11 at the smallest spatial scale to 0.47 at the largest scale. Consequently, we conducted all subsequent analyses using the largest scales: 16×16 m pixels for the sapling cover index, 30 m radius for mature beech basal area ([Fig. 1](#)) and a 15 m radius for all species basal area.

3.3. The effect of mature beech trees on beech saplings

The strength of the relationship between the beech sapling cover

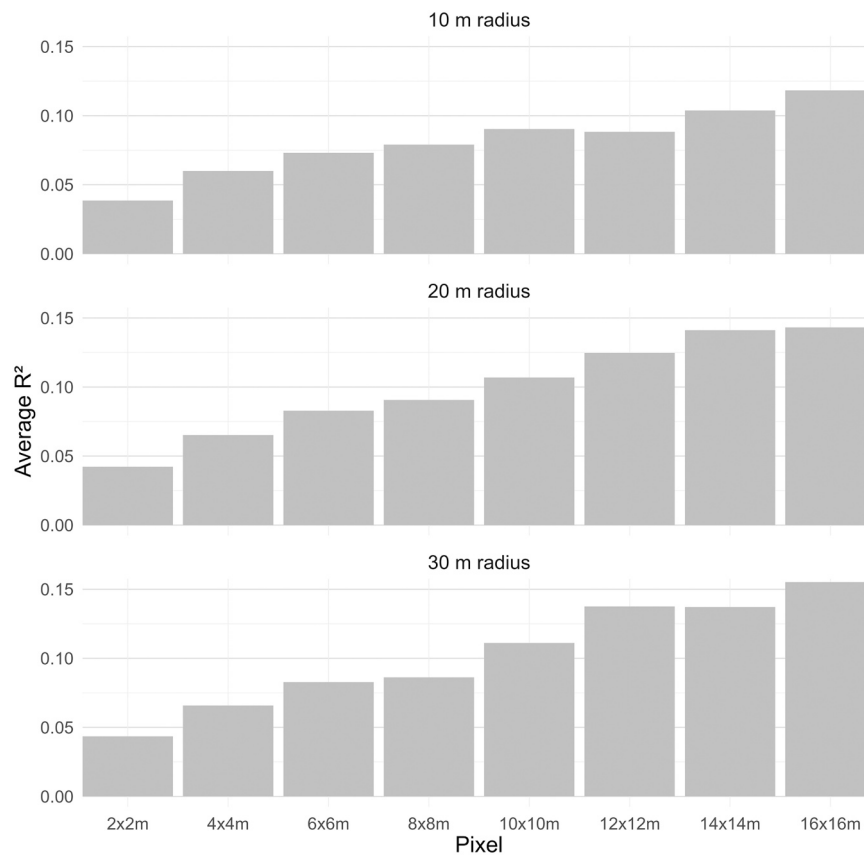


Fig. 2. Comparison of the average R^2 of the relationship between beech sapling cover index and beech basal area for each different radius at which beech basal area was calculated and at each different pixel at which beech sapling cover index was calculated, for all sites combined. Beech basal area considers all standing beech trees (DBH ≥ 15 cm).

Table 2

Characteristics of 11 stands scanned in Kenauk (sites 2, 4, 5, 6 and 13) forest and Gagnon Lake (sites 8, 9, 10, 11, 12 and 14) (Québec, Canada). All data in this table was acquired and extracted via mobile LiDAR scanning. The values in parentheses are standard deviation of the mean.

Site #	Basal area ($\text{m}^2 \cdot \text{ha}^{-1}$)	Mean DBH (cm)	Beech basal area ($\text{m}^2 \cdot \text{ha}^{-1}$)	Mean Beech DBH (cm)	Dead beech basal area ($\text{m}^2 \cdot \text{ha}^{-1}$)	Beech saplings basal area ($\text{m}^2 \cdot \text{ha}^{-1}$)
2	18.6	17.0 (0.08)	3.37	18.7 (0.06)	0.31	2.50
4	23.2	17.9 (0.10)	1.22	19.3 (0.07)	0	1.00
5	22.6	17.8 (0.10)	6.29	19.3 (0.06)	0.20	2.37
6	19.6	16.0 (0.07)	6.44	18.6 (0.06)	0.08	1.59
8	15.3	20.3 (0.09)	5.99	26.0 (0.07)	2.61	5.20
9	12.7	16.4 (0.08)	3.13	19.9 (0.06)	0.17	1.49
10	20.8	19.3 (0.08)	3.92	22.9 (0.07)	1.86	1.90
11	15.6	16.6 (0.08)	6.17	23.6 (0.08)	1.04	2.47
12	16.0	24.9 (1.40)	7.69	24.0 (0.07)	3.00	2.19
13	19.5	16.1 (0.06)	4.21	17.6 (0.04)	0.59	2.59
14	20.9	21.4 (0.09)	2.70	22.4 (0.08)	0.86	2.34

index and mature beech basal area varied across sites. A positive

relationship with an $R^2 \geq 0.10$ was observed in only four of our eleven sites (Fig. 4). Notably, seven of the eleven sites showed little to no relationship ($R^2 \leq 0.10$), indicating that this association was generally weak or absent across most stands. Among the four sites showing positive relationships, sites 4 and 9 exhibited relatively strong relationships ($R^2 = 0.47$), while weaker relationships were observed in sites 10 ($R^2 = 0.37$) and 8 ($R^2 = 0.17$). In regeneration ecology and spatial analyses, low to moderate R^2 values are expected because recruitment patterns are shaped by high environmental variability and multiple interacting drivers, and similar effect sizes are frequently reported in ecological studies (Lin and Wiegand, 2023). Therefore, R^2 values in the range of ~ 0.3 – 0.5 , such as those observed here, are often interpreted as moderate to strong depending on context. When analyzing the relationship between mature beech basal area and beech sapling basal area (instead of the sapling cover index), similar patterns emerged. Three of the same sites exhibited positive significant relationships, with slight variations in R^2 values, and a total of four sites showing significant relationships (Fig. 5). However, site 13 demonstrated a significant but weaker relationship ($R^2 = 0.14$), while site 8 showed no significant relationship when using beech sapling basal area. This discrepancy may be attributed to seasonal differences, as site 13 was scanned in autumn under leaf-off conditions, potentially making sapling basal area a more accurate indicator of sapling presence compared to the cover index, which relies on leaf cover. The number of saplings was also tested as a dependent variable in preliminary analyses, but it consistently resulted in weaker and less significant relationships than sapling basal area.

We acknowledge that not accounting for spatial autocorrelation (see Section 2.5) may have increased the apparent level of statistical significance (p-values) in some sites, as neighboring pixels within a stand are not fully independent. Conversely, the relatively small sample size at the

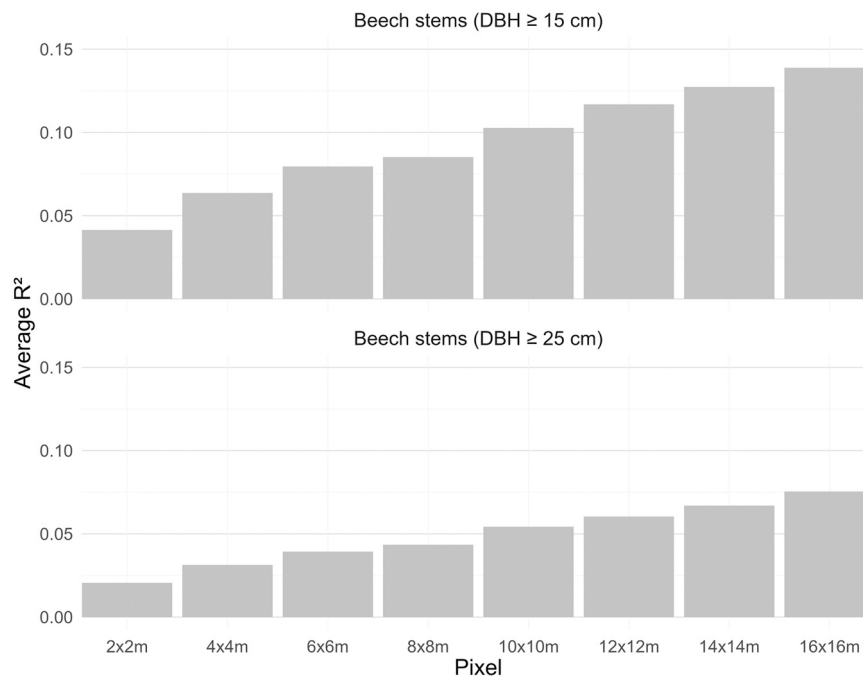


Fig. 3. Comparison of the average R^2 of the relationship between beech sapling cover index and beech basal area for all beech stems identified ($\text{DBH} \geq 15$ cm) and mature beech stems ($\text{DBH} \geq 25$ cm), at each different pixel at which beech sapling cover index was calculated, for all sites combined. Beech basal area considers all standing beech trees.

stand level (31–36 pixels per site) may have limited the ability to detect moderate relationships, thereby reducing the likelihood of obtaining higher R^2 values in some cases. Importantly, these two factors act in opposite directions. If spatial autocorrelation had been a dominant issue, a greater number of sites might be expected to exhibit significant relationships; however, only 4 out of 11 sites did so. In contrast, if limited sample size had been the primary constraint, detecting these significant relationships may have been less likely. Taken together, these elements suggest that the observed patterns are not driven by methodological issues.

3.4. The effect of canopy opening (total stand basal area) on beech saplings

The strength of the relationship between the beech sapling cover index and total stand basal area also showed high variability across sites, with significant relationships found in three over eleven sites (Fig. 6). Two sites (10 and 12; $R^2 = 0.31$ and $R^2 = 0.18$ respectively) exhibited significant negative relationships while site 9 showed a positive relationship ($R^2 = 0.30$). The remaining nine sites had insignificant relationships.

4. Discussion

In this study, we investigated the within-stand relationships between beech sapling cover as the dependent variable on one hand and beech mature tree density and canopy opening as the independent variables on the other hand. These relationships were thoroughly analyzed at various spatial scales for both dependant and independent variables. Contrary to our initial expectations based on existing literature, neither mature beech density nor canopy gaps consistently explained within-stand beech sapling cover, regardless of the spatial scale considered.

4.1. Relationship between beech sapling cover and mature beech basal area

The best relationship between beech sapling cover and mature beech

tree basal area (BA) was observed when beech sapling cover was evaluated in a 16 m pixel and mature beech density was evaluated in 30-m radius. However, this relationship varied greatly among stands and was present in only four of the eleven stands. This low occurrence (four out of eleven sites) is counterintuitive because we would have expected a more abundant regeneration around parent trees as several studies showed higher beech regeneration in stands with higher mature tree beech density (Bose et al., 2017; Nolet et al., 2008; Roy and Nolet, 2018).

We believe that one of the reasons to explain this low occurring within stand relationship relies in the beech capacity to regenerate both sexually and vegetatively. Most studies regarding beech regeneration presupposed that beech sapling domination in the understory originated from root suckers that developed after a major or partial canopy disturbance (Beaudet et al., 2007; Fajvan et al., 2019) or mortality of mature beech trees, potentially by BBD, that created thickets of clonal beech saplings (Cale et al., 2012; Jones and Raynal, 1987). A study in New York State found that minor changes in stand understory composition occurred after cutting in five uneven-aged stands where beech was either absent or very dense, leading them to conclude that beech regeneration was more likely to develop in proximity to a mature beech tree from root suckers and not take over new stands (Bohn and Nyland, 2003).

However, if beech dominance in the understory were mostly related to root suckers, we would expect that beech sapling cover be strongly positively related to mature beech basal area within a stand because of the rather limited radius of influence of this regeneration process (Held, 1983; Jones and Raynal, 1986). Beech sexual regeneration radius of influence is clearly wider because beech nuts can be dispersed by barochory usually up to 20 m away from the parent trees (Wagner et al., 2010). The radius is further augmented by small mammals like squirrels and birds like the blue jay (*Cyanocitta cristata*), who can disperse an average of 7 nuts at distances of 4 km (Johnson and Adkisson, 1985). Our results lead us indirectly to believe that sexual reproduction appears to have a stronger role to play in the establishment of beech dominance in the understory than previously thought, by dispersing beech regeneration further and more randomly. Accordingly, Dumont et al. (2023)

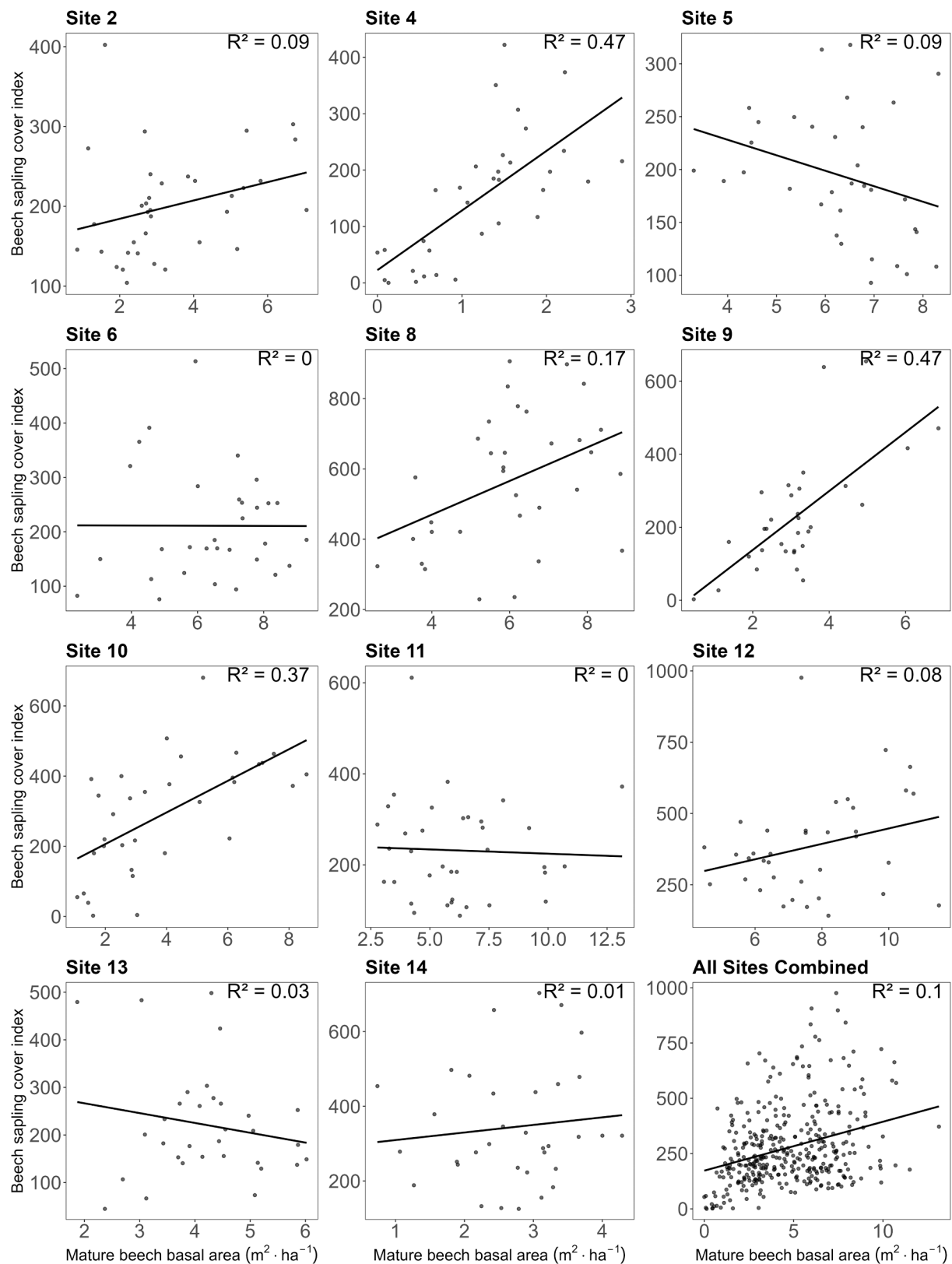


Fig. 4. Relationship between beech sapling cover index and mature beech basal area for each individual site. Beech sapling cover index was measured in a 16 m pixel while mature beech basal area was calculated in a 30 m radius centered on the pixels. See the methods [Section 2.4.2](#) for details concerning the calculation of the beech sapling cover index. Mature beech basal area considers all standing dead and alive beech trees ($\text{DBH} \geq 15 \text{ cm}$). The range of values of each axis varies among sites to improve readability and ensure that site-specific patterns and variability were clearly visible within the multi-panel layout; using fixed axis limits would have compressed the data in several panels and obscured relationships.

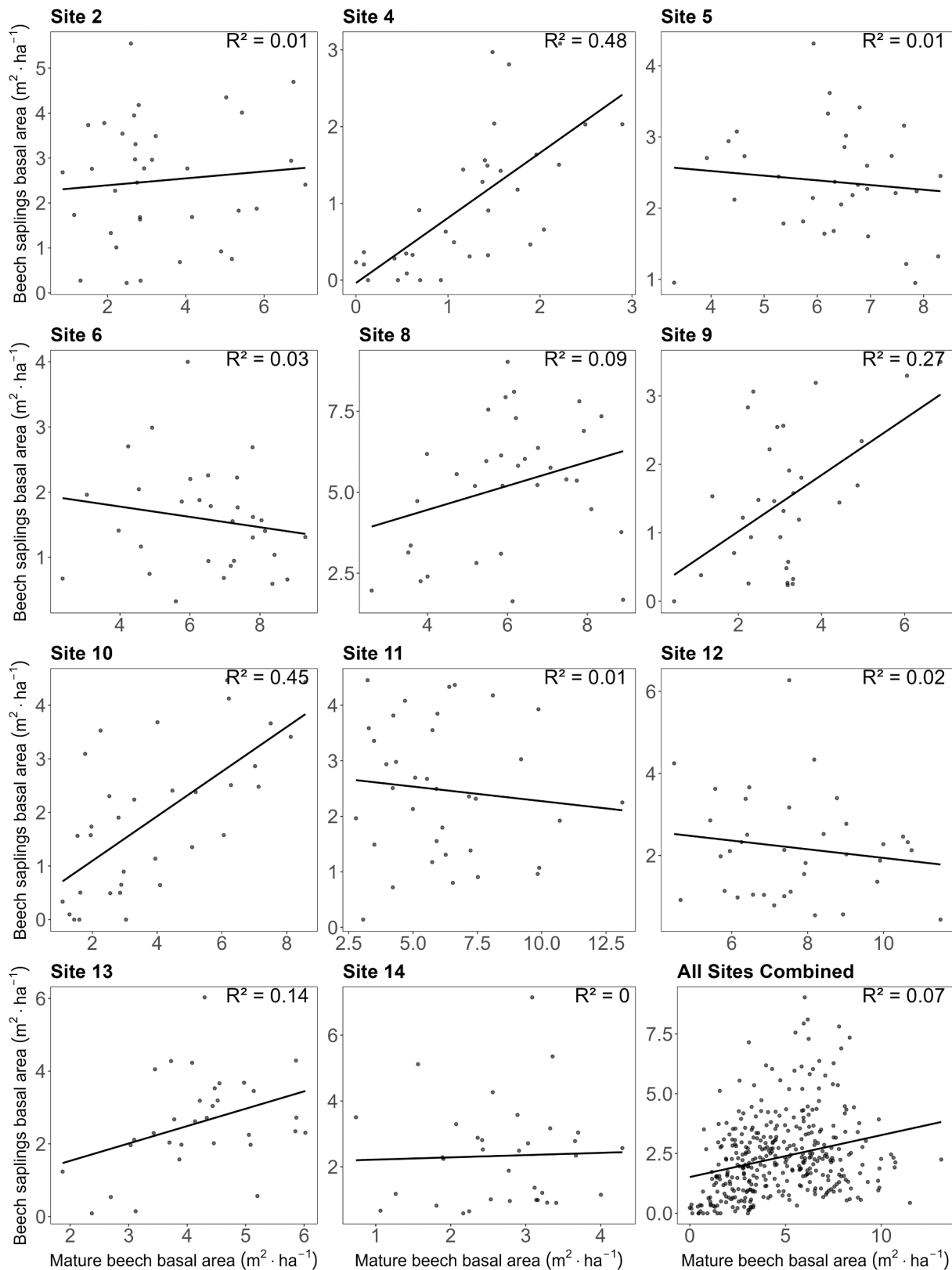


Fig. 5. Relationship between beech sapling basal area and mature beech basal area for each individual site. Beech sapling basal area was measured in a 16 m pixel while mature beech basal area was calculated in a 30 m radius centered on the pixels. See the methods [Section 2.4.3](#) for details concerning the calculation of the beech sapling basal area. Mature beech basal area considers all standing dead and alive beech trees ($\text{DBH} \geq 15$ cm). The range of values of each axis varies among sites to improve readability and ensure that site-specific patterns and variability were clearly visible within the multi-panel layout; using fixed axis limits would have compressed the data in several panels and obscured relationships.

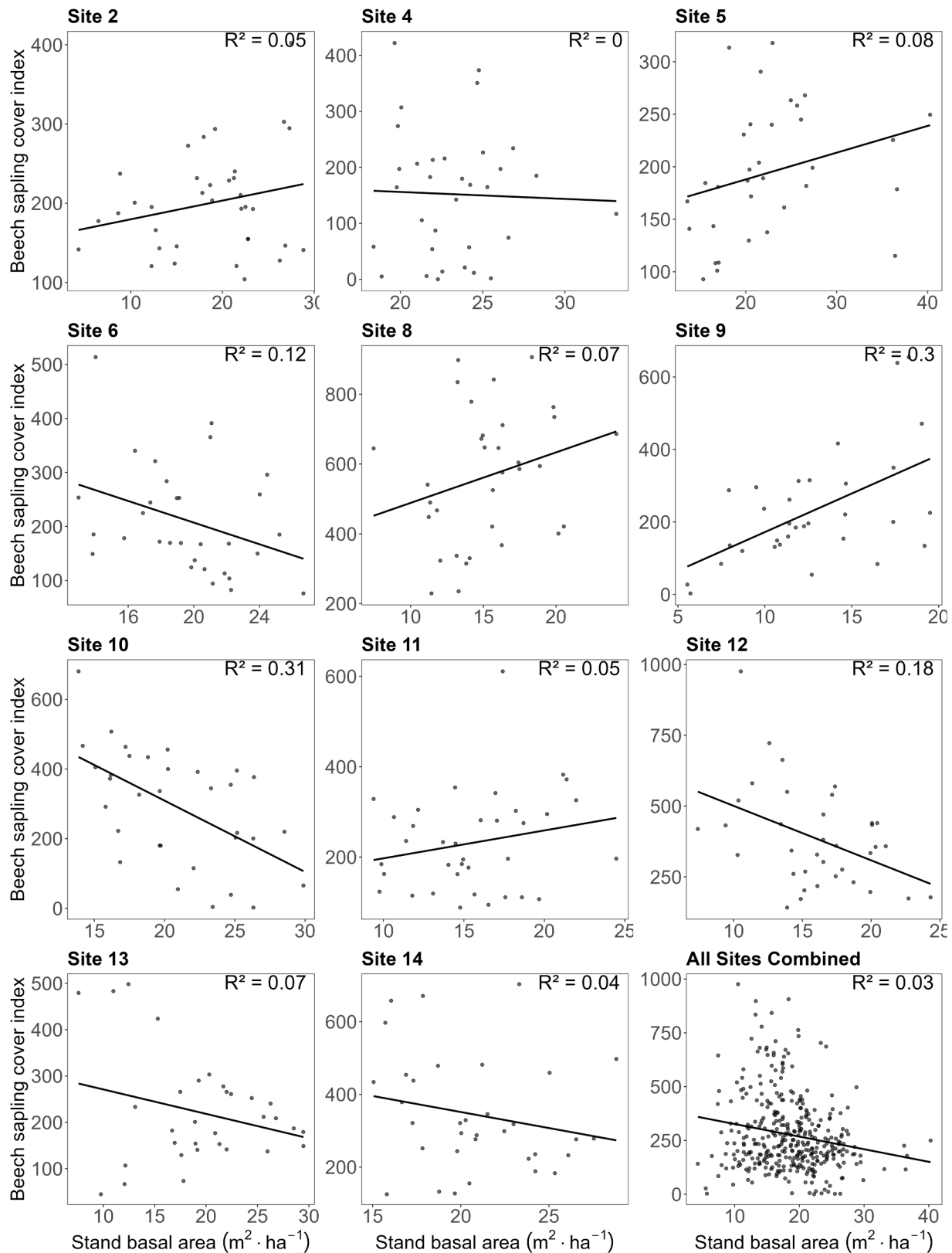


Fig. 6. Relationship between beech sapling cover index and stand basal area for each individual site. Beech sapling cover index was measured in a 16 m pixel while stand basal area was calculated in a 15 m radius centered on the pixels. See the methods [Section 2.4.2](#) for details concerning the calculation of the beech sapling cover index. Stand basal area considers all living trees ($\text{DBH} \geq 9$ cm). The range of values of each axis varies among sites to improve readability and ensure that site-specific patterns and variability were clearly visible within the multi-panel layout; using fixed axis limits would have compressed the data in several panels and obscured relationships.

found that beech seedlings (sexual origin) are a critical component of beech regeneration especially in stands with no or light harvesting, supporting the idea of the importance of sexual regeneration in the beech proliferation process.

4.2. Relationship between beech saplings and canopy opening

Several studies have shown a positive relationship between canopy opening and beech sapling abundance (Dumont et al., 2023). In the same vein, other studies demonstrated that beech saplings react strongly and positively to an increase in light availability (e.g. Nolet et al., 2015) and that beech saplings outgrow sugar maple when stands are partially harvested regardless of the intensity (Leduc et al., 2024). For these reasons, we expected to observe higher beech sapling cover where total stand basal area was lower. However, we found very little to no effect of the total stand basal area in relation to beech saplings throughout the 11 1-ha sites scanned in our study. Hence, our prediction that beech sapling cover would be negatively related to basal area within stands is not substantiated by our results. This is also surprising as it contrasts with other findings who observed a correlation between overstory basal area and beech sapling dominance and development (Bose et al., 2017; Nolet et al., 2008; Roy and Nolet, 2018).

One reason to explain this result might be that, at the fine scale at which we performed our analyses, stand basal area may fluctuate strongly and rapidly because of tree mortality. Hence it is possible that some pixels have experienced a very recent mortality while mortality occurred several years ago in other pixels. Consequently, the beech sapling layer had less time to benefit from higher light intensity due to mortality when this mortality was more recent. Our sampling could not provide such details about stand dynamics. We also acknowledge that in this study, canopy openness was inferred indirectly using stand basal area as a structural proxy for light availability. However, basal area provides only a coarse estimate of canopy conditions and may not have fully captured the fine-scale variability in light conditions that could have influenced sapling establishment and growth. Overall, observing that beech sapling dominance may develop within stands no matter the abundance of parent trees (and no matter the canopy opening level) let us believe that once beech starts to establish in a sugar maple stand, its dominance could be an ineluctable phenomenon, if no silvicultural means are deployed.

4.3. The importance of scale to describe regeneration

Our primary finding, that is the absence of a relationship between the beech sapling cover index and both canopy openness and mature beech BA, is reinforced by the fact that this absence was verified across multiple spatial scales. In contrast, a study conducted at a predefined spatial scale might have cast doubt on the validity of these findings. This is particularly relevant given that our results contradict expectations and findings reported in other studies employing different field data collection methods (Nolet et al., 2015). Recognizing that the optimal scale for detecting spatial relationships is rarely known, we adopted a methodology that allowed for the analysis of entire hectares of stands, enabling us to systematically adjust the scale of all calculated variables. This approach provided a unique advantage for capturing spatial variability.

Our analysis demonstrates that relationships observed when pooling plots from multiple stands and analyzing them collectively does not persist when a single stand (1 ha) is analyzed as an independent dataset. This contrast may partly reflect the modifiable areal unit problem (MAUP), as pooling data across stands combines within-stand and between-stand variability, potentially inflating or obscuring observed relationships. While no strong relationship between our variables was detected at any scale, a notable finding is that the strength of the relationship between mature beech BA and beech sapling cover increased with the scale at which these variables were calculated (Fig. 2). This

trend was consistent across all variables examined, suggesting that increasing the spatial scale of calculation consistently enhances the strength of observed relationships.

These findings call into question the methodologies traditionally employed to assess regeneration, particularly regarding the scale at which regeneration plots are defined and analyzed. Our results highlight limitations in combining small, fragmentary plots and extrapolating their findings to the stand level. Such plots may not adequately represent the variability within stands, as their size (Lombardi et al., 2015) or shape (Corona et al., 1998) can significantly influence stand characterization. Additionally, the use of isolated microplots to study regeneration dynamics may obscure the mechanisms operating at both fine and large spatial scales within stands. When analyzing entire stands, relationships between regeneration and basal area may differ from those derived from small, non-contiguous plots, emphasizing the need for methods such as MLS that better capture spatial heterogeneity at broader scales.

4.4. Novelty, benefits and caveats

While the use of MLS technology is relatively new in the forest domain, it is increasingly widely used (Di Stefano et al., 2021). It remains that most study emphasize on tree characteristics (tree detection, tree DBH, tree taper etc.) for various purposes such as on the detection of tree trunks and the calculation of their DBH (Chen et al., 2019; Liang et al., 2018; Marselis et al., 2016; Pierzchała et al., 2018; Shao et al., 2024), as well as comparing the performance of distinct types of LiDAR in these calculations (Bauwens et al., 2016; Bienert et al., 2018; Cabo et al., 2018; Choi and Song, 2022; Oveland et al., 2018). However, MLS has rarely (Nie et al., 2024) been used to describe the regeneration layer as it has been performed in this study because MLS inventories usually have difficulties detecting and calculating small DBH (< 4 cm) sapling trunks (Kuzelka et al., 2022).

Moreover, most previous studies have focused on less complex scanning environments, such as even-aged forests with large-diameter stems and minimal obstruction from the shrubs and saplings. In contrast, this study demonstrated that MLS, combined with light pruning of branches along pre-scanning paths, can effectively map the shrub and sapling layer. Finally, we also overcame a common caveat of LiDAR-based approaches in forest inventory by combining species composition with our 3D-map database. We used geolocated 3D photography (after pre-identifying all mature beech trees within the site manually) in combination with our 3D map to correctly identify all beech trees (DBH ≥ 15 cm) within each site. Ongoing research should lead soon to automatic species identification in combination with MLS as species composition is paramount in forest inventories.

When verified, our algorithm achieved very high accuracy, detecting 97% of stems and 98% of basal area (626 of 645 beech trees ≥ 15 cm DBH), with missed individuals largely limited to small, steeply inclined trees. The use of MLS allowed us to perform a comprehensive full tree inventory (totaling 8455 trees (DBH ≥ 9 cm)) across all study sites over a hectare in significantly less time compared to traditional manual field inventories. For instance, in a plantation with minimal understory vegetation, MLS was approximately four times faster than conventional field inventories for the same area (Ko et al., 2021). In the complex forest conditions we studied, we believe MLS provided even greater time savings. Furthermore, the application of MLS, combined with our analytical algorithms, detected 30,498 saplings (validated at over 91% accuracy through manual point-cloud verification) and allowed us to evaluate sapling layer cover across all sites, a level of detail unattainable with conventional sampling methods. The regeneration data collected through MLS also facilitated precise spatial mapping using LiDAR-generated 3D models, an accomplishment beyond the reach of traditional approaches. These advancements provide novel insights into the spatial scales at which regeneration dynamics should be investigated, underscoring the unique capabilities of MLS in enhancing

ecological research.

While LiDAR offers remarkable potential and utility for forest data acquisition, it is accompanied by several limitations and caveats. Although capable of scanning and mapping thousands of trees during an hour-long survey, successful scanning requires proper preparation of the terrain and path, which can be time-consuming. In our project, preparing the terrain and collecting the necessary data required approximately one day per hectare, primarily due to challenging field conditions. Furthermore, processing mobile terrestrial LiDAR data to extract field data from point clouds involves additional time and resources. Currently, data processing, extraction, and calculations cannot be fully automated, necessitating human verification for certain outputs generated by our algorithms. This reliance on manual oversight highlights the importance of further technological advancements to streamline workflows and reduce processing time.

4.5. Silviculture implications and possible solutions

The proliferation of beech in sugar maple-dominated stands has been examined from multiple perspectives, along with various strategies to mitigate its spread. While the findings of the present study alone offer limited silvicultural implications, they gain significance when contextualized within the broader body of literature on the topic. Drawing on both our findings and existing research, we propose potential silvicultural approaches to prevent or reverse beech dominance in sugar maple-dominated stands. Despite differences in management history between regions, sites within each cluster shared similar disturbance timelines, particularly in Gagnon Lake where all stands were harvested in 2006. The occurrence of both strong and weak relationships among sites with comparable disturbance histories suggests that time since disturbance does not appear to consistently mediate the spatial relationship between beech saplings and mature trees. The most important insight from our study is that stands currently exhibiting moderate beech sapling dominance (e.g., $2 \text{ m}^2\text{-ha}^{-1}$) are likely to progress toward complete beech sapling dominance (e.g., $5 \text{ m}^2\text{-ha}^{-1}$) due to ongoing beech seedling recruitment in the absence of silvicultural interventions. Hence, for stands with moderately to complete beech sapling dominance in the sapling layer, tree-selection harvesting (or other forms of partial harvesting) is not a viable solution, as it tends to promote the growth of beech saplings (Leduc et al., 2024). The complete removal of beech saplings remains a potential management option to promote sugar maple regeneration. However, this approach is costly, its success uncertain, and it may require repeated implementation during each forest rotation cycle.

Clear-cutting presents a potential solution to reverse beech sapling dominance, particularly if saplings are eliminated during the harvesting process (Nolet et al., 2015). However, the limited social acceptability of clear-cutting, alongside its ecological impacts (Barrette et al., 2024; Nolet et al., 2018), constrains its widespread application. Recently, a revised patch-cutting system was proposed that enables spatial and temporal planning of precommercial silvicultural treatments (Nolet et al., 2024). This system appears promising for stands dominated by beech in the sapling layer, as it allows for small, targeted clear-cuts (patches) that remove the beech sapling layer, coupled with targeted cleaning treatments to prevent regrowth in these areas.

For stands with limited beech sapling presence (e.g., $< 2 \text{ m}^2\text{-ha}^{-1}$), the risk of beech dominance in the sapling layer appears lower over the next few decades. For these stands, we recommend winter partial harvesting, focusing on mature beech trees to minimize root damage and reduce root suckering, as well as sexual regeneration. Intensive monitoring should accompany this approach to enable rapid elimination of emerging beech regeneration.

5. Conclusion

Using mobile laser scanning (MLS), this study evaluated within-stand

relationships between American beech sapling cover and forest structural attributes through spatially explicit analyses conducted across entire 1 ha forest stands. Our results first highlight the strong potential of MLS for spatially continuous, stand-scale assessments of forest regeneration, while also emphasizing the need to better integrate species composition information into MLS-based approaches. Second, the absence of clear relationships between beech sapling cover, mature beech basal area, and canopy openness suggests that the relative contribution of sexual regeneration to the beech proliferation phenomenon may be greater than previously anticipated, compared with earlier assumptions that emphasized vegetative reproduction. Understanding how a species regenerates is fundamental in forest ecology and silviculture, as regeneration pathways strongly influence management strategies aimed at promoting or suppressing a given species. Given the ecological and silvicultural importance of beech proliferation in north-eastern North America sugar maple stands, further research, particularly genetic analyses, will be required to clarify the relative contributions of sexual and vegetative regeneration.

CRediT authorship contribution statement

Philippe Nolet: Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Pascal Rochon:** Software, Methodology, Data curation. **Angélique Dupuch:** Writing – review & editing, Formal analysis. **Frédéric Moore:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Philippe Nolet reports financial support was provided by Natural Sciences and Engineering Research Council of Canada. Philippe Nolet reports financial support was provided by Ministry of Natural Resources and Forests. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The dataset used is cited in the manuscript and attached in the submission files. Custom algorithms, scripts and raw LiDAR point clouds can be made available on request.

Mendeley Data

[Dataset - American beech sapling dominance: stand-scale insights from mobile laser scanning](#)

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Glossary

LiDAR: Light detection and ranging
BBD: Beech bark disease
MLS: Mobile laser scanning
DBH: Diameter at breast height
BA: Basal area
ALB: Augmented LiDAR box
SLAM: Simultaneous localization and mapping
TIN: Triangulated irregular network